

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

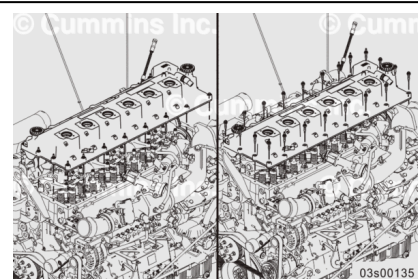
Additional Service Items

- No additional service items required.

General Information

Two types of rocker lever covers are available with different noise, vibration, and harshness performance.

For the rocker lever cover with noise reduction function, the mounting capscrews and compression limiters are part of the rocker lever cover. The capscrews are the captive design and will be held in place in the rocker lever cover.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

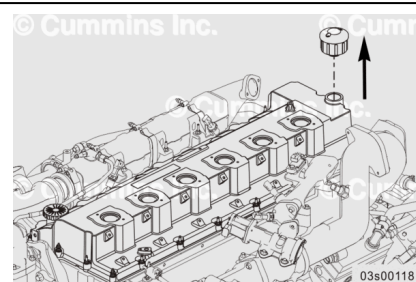
- Remove the Exhaust Gas Recirculation (EGR) crossover tube above the rocker lever cover. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html>)
- Disconnect the turbocharger actuator air line from the rocker lever cover. Refer to Procedure 010-118 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-118.html>)

- Disconnect the wiring harness on the rocker lever cover. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html>)
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the ignition coils. Refer to Procedure 013-012 in Section 13. (</qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html>)
- Remove the crankcase breather tube from rocker lever cover, if equipped. Refer to Procedure 003-024 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-024-tr.html>)

Remove

Accumulation of lubricating oil in the pressure relief valve when the temperature is below 0°C [32°F] is normal.

Remove the pressure relief valve from the rocker lever cover slowly and vertically, if equipped.

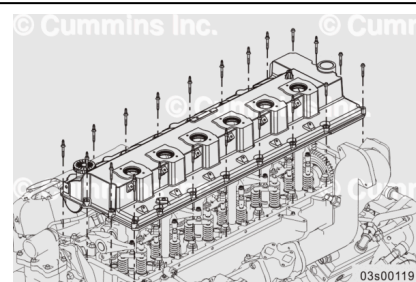


For the rocker lever cover without noise reduction function:

Remove the capscrews.

Remove the cover by lifting straight up.

The rocker lever cover gasket is reusable. Do **not** damage the gasket when removing or cleaning.

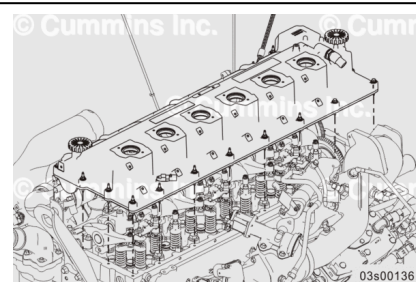


For the rocker lever cover with noise reduction function:

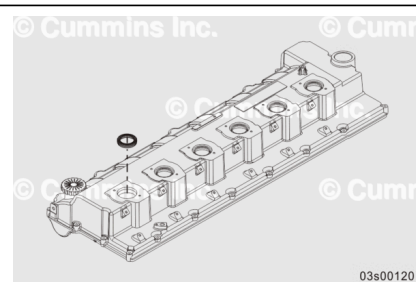
Loosen the rocker lever cover capscrews.

Loosen the brackets that secure the wiring harness plugs to the valve cover.

Pull the rocker lever cover gently. Leave the wiring harness plugs on the cylinder head.



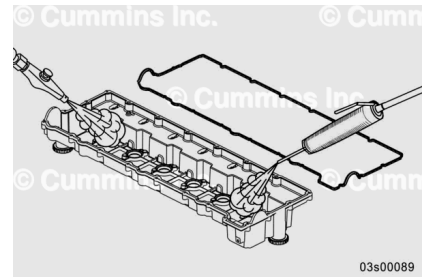
Remove the dust seal from the rocker lever cover, if required.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Disassemble the gasket from the rocker lever cover.

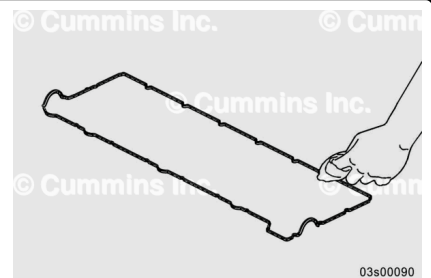
Steam clean the rocker lever cover. Verify the gasket groove is clean and free of debris.

White, tan, or brown foam or froth is sometimes found inside the rocker lever cover. Foam or froth can occur if the engine operates below thermostat opening temperature for sustained periods. This type of foam or froth will **not** affect the function of the engine. Foam or froth can also occur if there is an internal coolant leak. If an internal coolant leak is suspected, perform the Coolant Leak - Internal troubleshooting symptom tree.

Dry with compressed air.

⚠ CAUTION ⚠

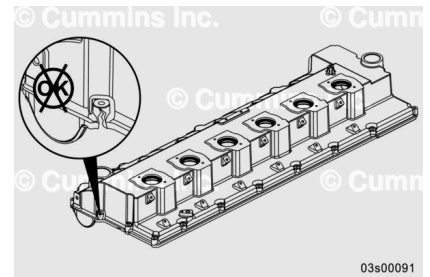
Do not use solvent to clean the rocker lever cover gasket. Solvent will damage the gasket material and cause it to swell. Prolonged exposure can also cause degradation of the gasket material.



Clean the reusable gasket with a dry, lint-free cloth, making sure no oil remains between the sealing beads.

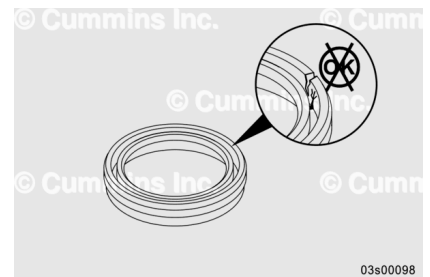
Inspect the rocker lever cover for cracks and other damage.

Replace the cover, if necessary.



Inspect the dust seal for damage.

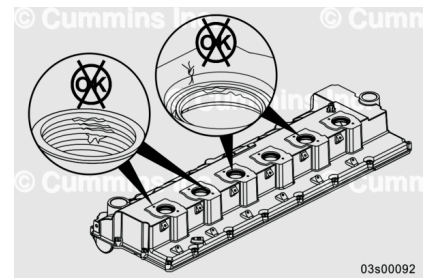
Replace the dust seal if damaged.



Inspect the spark plug tube seals for cracks, tears, or brittleness.

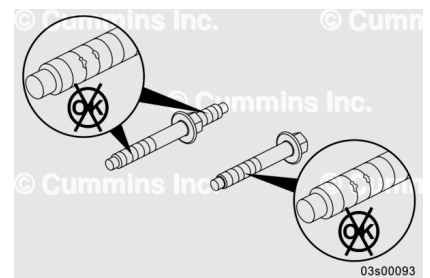
Inspect the threaded coil mounting holes for damage.

Replace the valve cover if damage is found.



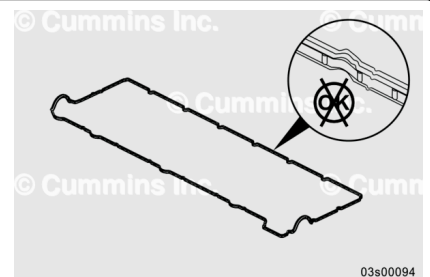
Inspect the capscrews for damage.

Replace the capscrews if damage is found.



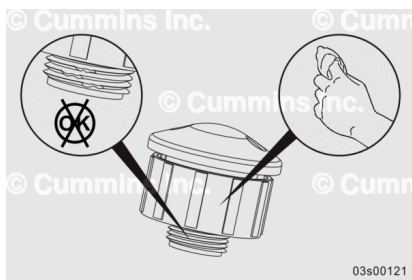
Inspect the rocker cover gasket beads for cuts, missing beads, kinks, twists, wear, and other damage.

Replace the gasket, if necessary.



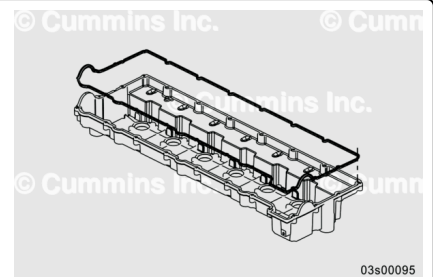
Clean the pressure relief valve if lubricating oil is found.

Replace the pressure relief valve if damage is found.



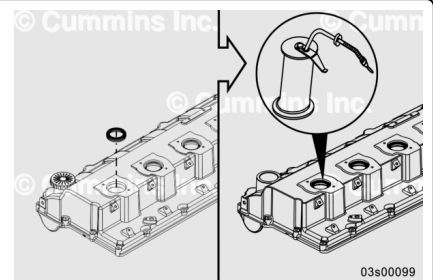
Install

If **not** damaged, the gasket can be used again. If damaged, the gasket **must** be discarded and a new one installed.



If the dust seal is removed, press the dust seal into the rocker lever cover. Press to stop.

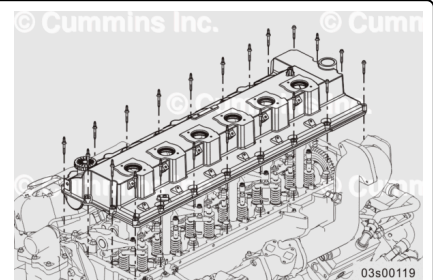
Lubricate the inner circumference of the dust seal with clean engine oil.



Install the rocker lever cover onto the rocker lever housing.

Align the coil mounting holes in the rocker lever cover with the spark plug tubes in the rocker lever housing. The cover **must** be gently pressed until the tubes fit through the seals prior to tightening the capscrews.

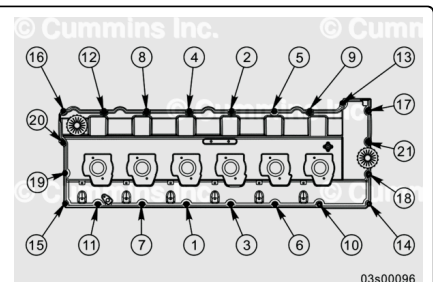
Loosely hand tighten the capscrews.



Although different in appearance, the relative location and the capscrew number remain the same.

For the rocker lever cover without noise reduction function:

Tighten capscrews 1 through 4 in the sequence shown in the illustration.



Torque Value: 5 n·m [44 in-lb]

Tighten all the capscrews in the sequence shown in the illustration.

Torque Value: 10 n•m [89 in-lb]

For the rocker lever cover with noise reduction function:

Tighten capscrews 1 through 4 in the sequence shown in the illustration.

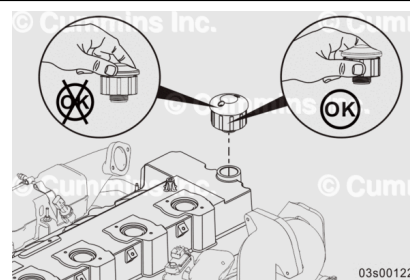
Torque Value: 5 n•m [44 in-lb]

Tighten all the capscrews in the sequence shown in the illustration.

Torque Value: 12 n•m [106 in-lb]

Install the pressure relief valve onto the rocker lever cover, if equipped.

Do **not** tighten the pressure relief valve cover. Tighten the seat of the pressure relief valve.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the crankcase breather tube, if removed. Refer to Procedure 003-024 in Section 3. (</qs3/pubsys2/xml/en/procedures/1016/1016-003-024-tr.html>)
- Install the ignition coils. Refer to Procedure 013-012 in Section 13. (</qs3/pubsys2/xml/en/procedures/1016/1016-013-012.html>)
- Connect the wiring harness on the rocker lever cover. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/1016/1016-019-043.html>)
- Connect the turbocharger actuator air line on the rocker lever cover. Refer to Procedure 010-118 in Section 10. (</qs3/pubsys2/xml/en/procedures/1016/1016-010-118.html>)
- Install the EGR crossover tube above the rocker lever cover. Refer to Procedure 011-070 in Section 11. (</qs3/pubsys2/xml/en/procedures/1016/1016-011-070-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for proper operation and leaks.

Last Modified: 11-Nov-2024
